



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

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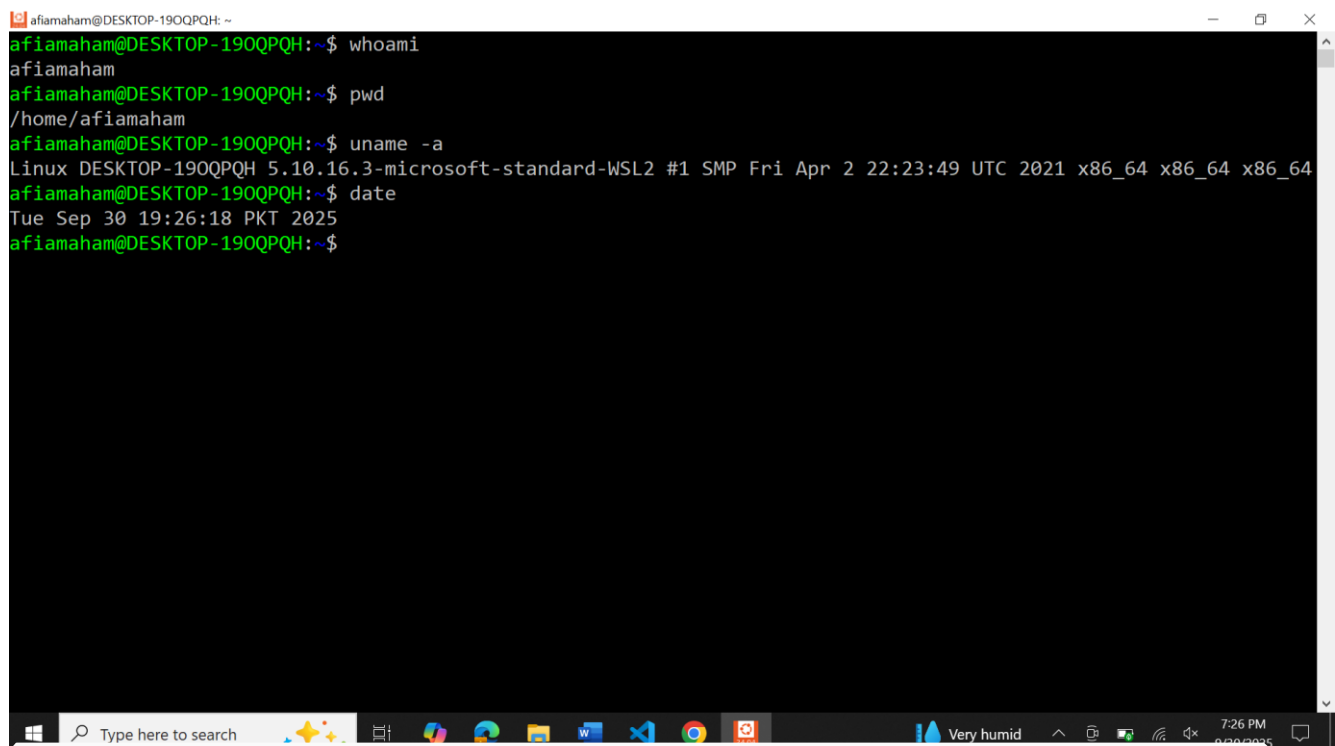
Lab 2: Linux Basics and Introduction

Part 1: Linux Environment Orientation

1.1 Understanding the Linux Environment

- ♦ **Concepts to Cover:**
 - What is Linux? Brief history and distributions
 - Linux vs Windows: Key differences
 - Understanding the shell (bash)
 - WSL2 as a Linux environment
- ♦ **Hands-on Activity:**

```
# Students open WSL2 terminal and explore
whoami          # Check current user
pwd             # Print working directory
uname -a        # System information
date            # Current date and time
```



The screenshot shows a Windows terminal window titled "afiamaham@DESKTOP-190QPQH: ~". The terminal output is as follows:

```
afiamaham@DESKTOP-190QPQH:~$ whoami
afiamaham
afiamaham@DESKTOP-190QPQH:~$ pwd
/home/afiamaham
afiamaham@DESKTOP-190QPQH:~$ uname -a
Linux DESKTOP-190QPQH 5.10.16.3-microsoft-standard-WSL2 #1 SMP Fri Apr 2 22:23:49 UTC 2021 x86_64 x86_64 x86_64
afiamaham@DESKTOP-190QPQH:~$ date
Tue Sep 30 19:26:18 PKT 2025
afiamaham@DESKTOP-190QPQH:~$
```

The Windows taskbar is visible at the bottom, showing the search bar, task view button, and several application icons. The system tray on the right indicates "Very humid" and the time "7:26 PM 9/30/2025".

- 1.2
- ♦

```
man ls
```

 # Manual pages
 - ♦

```
ls --help
```

 # Built-in help
 - ♦

```
which ls
```

 # Location of commands
 - ♦

```
type ls
```

 # Command type information

```
afiamaham@DESKTOP-19OQPQH: ~
LS(1) User Commands LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file

    -b, --escape
        print C-style escapes for nongraphic characters

Manual page ls(1) line 1 (press h for help or q to quit)
```

```
afiamaham@DESKTOP-19OQPQH: ~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all
        do not ignore entries starting with .
  -A, --almost-all
        do not list implied . and ..
  --author
        with -l, print the author of each file
  -b, --escape
        print C-style escapes for nongraphic characters
  --block-size=SIZE
        with -l, scale sizes by SIZE when printing them;
        e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups
        do not list implied entries ending with ~
  -c
        with -lt: sort by, and show, ctime (time of last
        change of file status information);
        with -l: show ctime and sort by name;
        otherwise: sort by ctime, newest first

  -C
        list entries by columns
  --color[=WHEN]
        color the output WHEN; more info below
  -d, --directory
        list directories themselves, not their contents
  -D, --dired
        generate output designed for Emacs' dired mode
  -f
        list all entries in directory order
  -F, --classify[=WHEN]
        append indicator (one of */=>@|) to entries WHEN
        likewise, except do not append '*'
  --file-type
        across -x, commas -m, horizontal -x, long -l,
  --format=WORD
```

```
afiamaham@DESKTOP-19OQPQH: ~  
afiamaham@DESKTOP-19OQPQH:~$ which ls  
/usr/bin/ls  
afiamaham@DESKTOP-19OQPQH:~$ type ls  
ls is aliased to `ls --color=auto`  
afiamaham@DESKTOP-19OQPQH:~$
```

Part 2: File System Navigation

2.1 Understanding Linux Directory Structure

- ◆ Concepts to Cover:
 - Root directory (/)
 - Important directories: /home, /usr, /etc, /var, /tmp

- Absolute vs relative paths
- Hidden files and directories

• Demonstration:

```
ls /           # Root directory contents
ls -la        # Long listing with hidden files
cd /home      # Change directory
cd ~          # Home directory shortcut
cd -          # Previous directory
```

```
afiamaham@DESKTOP-19OQPQH: ~$ ls /
bin          dev          init          lib64        mnt          root         sbin.usr-is-merged  sys  var
bin.usr-is-merged  etc          lib           lost+found  opt          run          snap              tmp
boot         home        lib.usr-is-merged  media       proc         sbin        srv              usr
afiamaham@DESKTOP-19OQPQH: ~$ ls -la
total 52
drwxr-x--- 7 afiamaham afiamaham 4096 Sep 30 19:31 .
drwxr-xr-x 3 root      root      4096 Sep 29 19:33 ..
-rw-r----- 1 afiamaham afiamaham 275 Sep 29 19:54 .bash_history
-rw-r----- 1 afiamaham afiamaham 220 Sep 29 19:33 .bash_logout
-rw-r----- 1 afiamaham afiamaham 3771 Sep 29 19:33 .bashrc
drwxr-xr-x 3 afiamaham afiamaham 4096 Sep 29 19:50 .cache
-rw-r----- 1 afiamaham afiamaham 56 Sep 29 19:45 .gitconfig
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:33 .landscape
-rw-r----- 1 afiamaham afiamaham 20 Sep 30 19:31 .lesshtst
-rw-r----- 1 afiamaham afiamaham 0 Sep 30 19:21 .motd_shown
-rw-r----- 1 afiamaham afiamaham 807 Sep 29 19:33 .profile
drwx----- 2 afiamaham afiamaham 4096 Sep 29 19:48 .ssh
-rw-r----- 1 afiamaham afiamaham 0 Sep 29 19:33 .sudo_as_admin_successful
drwxr-xr-x 5 afiamaham afiamaham 4096 Sep 29 19:50 .vscode-server
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:51 lab01
afiamaham@DESKTOP-19OQPQH: ~$
```

```
afiamaham@DESKTOP-19OQPQH: /home$ cd /home
afiamaham@DESKTOP-19OQPQH: /home$ cd ~
afiamaham@DESKTOP-19OQPQH: ~$ cd -
/home
afiamaham@DESKTOP-19OQPQH: /home$
```

2.2 Basic Navigation Commands (15 minutes)

Commands to practice:

```
pwd          # Present working directory
ls           # List directory contents
ls -l        # Long format
ls -la       # Include hidden files
ls -lh       # Human readable sizes
cd           # Change directory
cd ..        # Parent directory
cd ~         # Home directory
cd /         # Root directory
```

```
afiamaham@DESKTOP-190QPQH: ~
afiamaham@DESKTOP-190QPQH:~$ pwd
/home/afiamaham
afiamaham@DESKTOP-190QPQH:~$ ls
lab01
afiamaham@DESKTOP-190QPQH:~$ ls -l
total 4
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:51 lab01
afiamaham@DESKTOP-190QPQH:~$ ls -la
total 52
drwxr-x--- 7 afiamaham afiamaham 4096 Sep 30 19:31 .
drwxr-xr-x 3 root      root      4096 Sep 29 19:33 ..
-rw----- 1 afiamaham afiamaham 414 Sep 30 19:39 .bash_history
-rw-r--r-- 1 afiamaham afiamaham 220 Sep 29 19:33 .bash_logout
-rw-r--r-- 1 afiamaham afiamaham 3771 Sep 29 19:33 .bashrc
drwxr-xr-x 3 afiamaham afiamaham 4096 Sep 29 19:50 .cache
-rw-r--r-- 1 afiamaham afiamaham 56 Sep 29 19:45 .gitconfig
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:33 .landscape
-rw----- 1 afiamaham afiamaham 20 Sep 30 19:31 .lessht
-rw-r--r-- 1 afiamaham afiamaham 0 Sep 30 19:21 .motd_shown
-rw-r--r-- 1 afiamaham afiamaham 807 Sep 29 19:33 .profile
drwx----- 2 afiamaham afiamaham 4096 Sep 29 19:48 .ssh
-rw-r--r-- 1 afiamaham afiamaham 0 Sep 29 19:33 .sudo_as_admin_successful
drwxr-xr-x 5 afiamaham afiamaham 4096 Sep 29 19:50 .vscode-server
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:51 lab01
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: /
afiamaham@DESKTOP-190QPQH:~$ ls -lh
total 4.0K
drwxr-xr-x 2 afiamaham afiamaham 4.0K Sep 29 19:51 lab01
afiamaham@DESKTOP-190QPQH:~$ cd
afiamaham@DESKTOP-190QPQH:~$ cd ..
afiamaham@DESKTOP-190QPQH:/home$ cd ~
afiamaham@DESKTOP-190QPQH:~$ cd /
afiamaham@DESKTOP-190QPQH:/$
```

Part 3: File and Directory Operations

**3.1 Creating and Managing Files/Directories

- ♦ Commands to demonstrate:

```
mkdir mylab2           # Create directory
mkdir -p test/sub/dir  # Create nested directories
touch file1.txt        # Create empty file
touch file2.txt file3.txt # Multiple files

# Text editors introduction
nano hello.txt         # Simple text editor
# OR
echo "Hello Linux!" > hello.txt # Redirect output to file
```

- ♦ File viewing commands:

```
cat hello.txt          # Display file contents
less hello.txt         # Page through file
head hello.txt         # First 10 lines
```

tail hello.txt
wc hello.txt

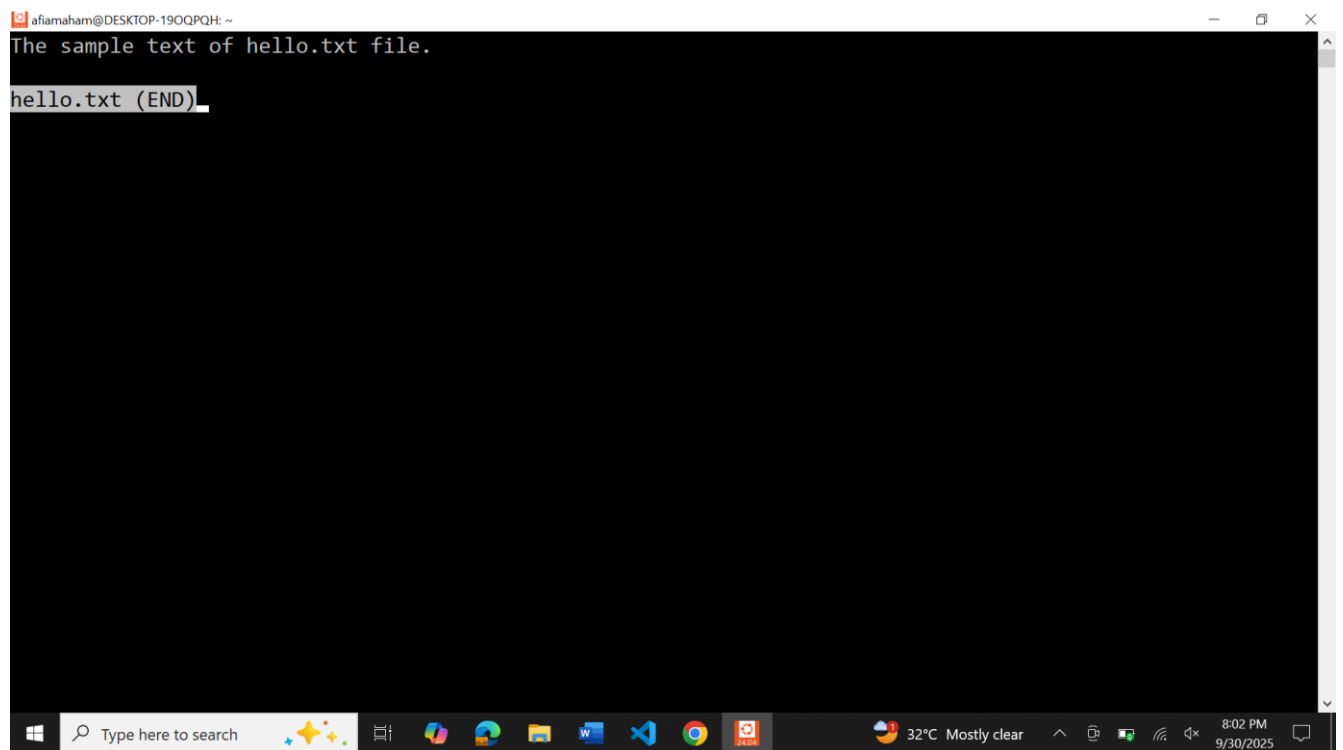
Last 10 lines
Word count

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ mkdir mylab2  
afiamaham@DESKTOP-190QPQH:~$ mkdir -p test/sub/dir  
afiamaham@DESKTOP-190QPQH:~$ touch file1.txt  
afiamaham@DESKTOP-190QPQH:~$ touch file2.txt file3.txt  
afiamaham@DESKTOP-190QPQH:~$ nano hello.txt  
afiamaham@DESKTOP-190QPQH:~$
```

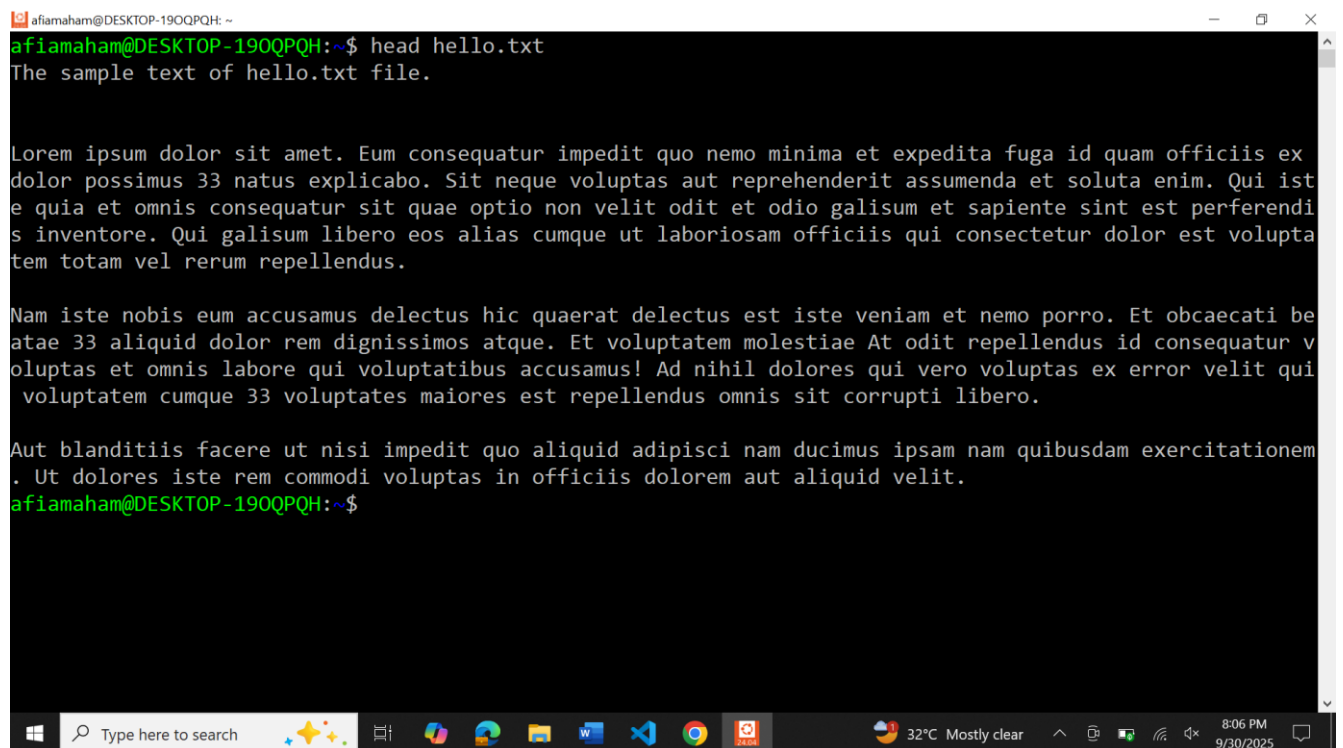
```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ cat hello.txt  
The sample text of hello.txt file.  
afiamaham@DESKTOP-190QPQH:~$
```



```
afiamaham@DESKTOP-19OQPQH: ~  
The sample text of hello.txt file.  
hello.txt (END)
```



```
afiamaham@DESKTOP-19OQPQH: ~$ head hello.txt  
The sample text of hello.txt file.  
  
Lorem ipsum dolor sit amet. Eum consequatur impedit quo nemo minima et expedita fuga id quam officiis ex  
dolor possimus 33 natus explicabo. Sit neque voluptas aut reprehenderit assumenda et soluta enim. Qui ist  
e quia et omnis consequatur sit quae optio non velit odit et odio galisum et sapiente sint est perferendi  
s inventore. Qui galisum libero eos alias cumque ut laboriosam officiis qui consectetur dolor est volupta  
tem totam vel rerum repellendus.  
  
Nam iste nobis eum accusamus delectus hic quaerat delectus est iste veniam et nemo porro. Et obcaecati be  
atae 33 aliquid dolor rem dignissimos atque. Et voluptatem molestiae At odit repellendus id consequatur v  
oluptas et omnis labore qui voluptatibus accusamus! Ad nihil dolores qui vero voluptas ex error velit qui  
voluptatem cumque 33 voluptates maiores est repellendus omnis sit corrupti libero.  
  
Aut blanditiis facere ut nisi impedit quo aliquid adipisci nam ducimus ipsam nam quibusdam exercitationem  
. Ut dolores iste rem commodi voluptas in officiis dolorem aut aliquid velit.  
afiamaham@DESKTOP-19OQPQH:~$
```



```
afiamaham@DESKTOP-19OQPQH: ~  
afiamaham@DESKTOP-19OQPQH:~$ tail hello.txt  
The sample text of hello.txt file.  
  
Lorem ipsum dolor sit amet. Eum consequatur impedit quo nemo minima et expedita fuga id quam officiis ex  
dolor possimus 33 natus explicabo. Sit neque voluptas aut reprehenderit assumenda et soluta enim. Qui ist  
e quia et omnis consequatur sit quae optio non velit odit et odio galisum et sapiente sint est perferendi  
s inventore. Qui galisum libero eos alias cumque ut laboriosam officiis qui consectetur dolor est volupta  
tem totam vel rerum repellendus.  
  
Nam iste nobis eum accusamus delectus hic quaerat delectus est iste veniam et nemo porro. Et obcaecati be  
atae 33 aliquid dolor rem dignissimos atque. Et voluptatem molestiae At odit repellendus id consequatur v  
oluptas et omnis labore qui voluptatibus accusamus! Ad nihil dolores qui vero voluptas ex error velit qui  
voluptatem cumque 33 voluptates maiores est repellendus omnis sit corrupti libero.  
  
Aut blanditiis facere ut nisi impedit quo aliquid adipisci nam ducimus ipsam nam quibusdam exercitationem  
. Ut dolores iste rem commodi voluptas in officiis dolorem aut aliquid velit.  
afiamaham@DESKTOP-19OQPQH:~$
```

```
afiamaham@DESKTOP-19OQPQH: ~  
afiamaham@DESKTOP-19OQPQH:~$ wc hello.txt  
 8 164 1074 hello.txt  
afiamaham@DESKTOP-19OQPQH:~$
```

3.2 Copying, Moving, and Deleting

- ◆ **Commands to practice:**

```
cp hello.txt backup.txt      # Copy file  
cp -r mylab2 mylab2_backup  # Copy directory recursively  
mv backup.txt renamed.txt   # Move/rename file  
rm renamed.txt              # Remove file  
rm -r mylab2_backup          # Remove directory  
rmdir empty_directory        # Remove empty directory
```

Hands-on Exercise: Students create a directory structure, add files, and practice file

operations.

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ cp hello.txt backup.txt  
afiamaham@DESKTOP-190QPQH:~$ cp -r mylab2 mylab2_backup  
afiamaham@DESKTOP-190QPQH:~$ mv backup.txt renamed.txt  
afiamaham@DESKTOP-190QPQH:~$ rm renamed.txt  
Command 'rm' not found, but can be installed with:  
sudo apt install remind  
afiamaham@DESKTOP-190QPQH:~$ rm renamed.txt  
afiamaham@DESKTOP-190QPQH:~$ rm -r mylab2_backup  
afiamaham@DESKTOP-190QPQH:~$ rmdir empty_directory  
rmdir: failed to remove 'empty_directory': No such file or directory  
afiamaham@DESKTOP-190QPQH:~$ rmdir mylab2  
afiamaham@DESKTOP-190QPQH:~$
```

Part 4: File Permissions and Ownership

4.1 Understanding File Permissions

- ◆ **Concepts to Cover:**
 - Permission types: read (r), write (w), execute (x)
 - Permission groups: user (u), group (g), others (o)
 - Numeric notation: 755, 644, etc.
- ◆ **Commands to demonstrate:**

```
ls -l           # View permissions  
chmod 755 file.txt # Change permissions (numeric)  
chmod u+x file.txt # Add execute permission for user  
chmod g-w file.txt # Remove write permission for group  
chown user:group file.txt # Change ownership (if applicable)
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ ls -l  
total 12  
-rw-r--r-- 1 afiamaham afiamaham 0 Sep 30 19:50 file1.txt  
-rw-r--r-- 1 afiamaham afiamaham 0 Sep 30 19:50 file2.txt  
-rw-r--r-- 1 afiamaham afiamaham 0 Sep 30 19:50 file3.txt  
-rw-r--r-- 1 afiamaham afiamaham 1074 Sep 30 20:05 hello.txt  
drwxr-xr-x 2 afiamaham afiamaham 4096 Sep 29 19:51 lab01  
drwxr-xr-x 3 afiamaham afiamaham 4096 Sep 30 19:49 test  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ chmod 755 file1.txt  
afiamaham@DESKTOP-190QPQH:~$ chmod u+x file1.txt  
afiamaham@DESKTOP-190QPQH:~$ chmod g-w file1.txt  
afiamaham@DESKTOP-190QPQH:~$
```

Part 5: Text Processing and Utilities

5.1 Essential Text Commands

- ◆ Commands to demonstrate:

```
grep "pattern" file.txt      # Search for patterns  
grep -i "pattern" file.txt   # Case-insensitive search  
grep -n "pattern" file.txt   # Show line numbers
```

5.2 Pipes and Redirection

- ◆ Concepts and commands:

```
ls -l | grep ".txt"          # Pipe output
cat file1.txt file2.txt > combined.txt # Redirect output
echo "new line" >> file.txt # Append to file
sort file.txt | uniq         # Chain commands
```

Part 6: Introduction to Processes

6.1 Understanding Processes

- ◆ Concepts to Cover:

- What is a process?
- Process ID (PID)
- Parent-child relationships
- Process states

- ◆ Commands to demonstrate:

```
ps                # Show current processes
ps aux            # Detailed process list
ps -ef           # Full format listing
pstree            # Process tree
top               # Real-time process viewer
htop              # Enhanced process viewer (if available)
kill PID          # Terminate process by PID
killall process_name # Kill processes by name
pkill pattern     # Kill processes matching pattern
```